RESEARCH ARTICLE

Use of Robust Norming to Create a Sensitive Cognitive Summary Score in *De Novo* Parkinson's Disease: An Illustrative Example

Daniel Weintraub, MD,[1,2*] Michael C. Brumm, MS,[3] Ryan Kurth, MS,[3] Michele K. York, PhD, ABPP-CN,[4] and the Parkinson's Progression Markers Initiative

ABSTRACT: Background: Cognitive impairment is common at all stages of Parkinson's disease (PD), but there is no consensus on which neuropsychological tests to use or how to interpret cognitive battery results. A cognitive summary score (CSS) combines the richness of a neuropsychological battery with the simplicity of a single score.

Objective: The objective of this study was to determine whether a CSS created using robust norming can detect early cognitive deficits in de novo, untreated PD.

Methods: Baseline cognitive data from PD participants and healthy control participants (HCs) in the Parkinson's Progression Markers Initiative were used to (1) create a robust HC subgroup without cognitive decline, (2) generate regression-based *z* scores for six cognitive measures using this subgroup, and (3) create a CSS by averaging all *z* scores.

Results: PD participants scored worse than HCs on all cognitive tests, with larger effects when compared with the robust HC subgroup rather than all HCs. Applying internally derived norms, the largest effects were for processing speed/working memory (Cohen's $d = -0.55$) and verbal episodic memory (Cohen's $d = -0.48$ and -0.52). Robust norming shifted PD performance from average (CSS *z* score $= -0.01$) to low average (CSS *z* score $= -0.40$), with a larger effect for the CSS (PD vs. robust HC subgroup; Cohen's $d = -0.60$) compared with individual tests.

Conclusions: Patients with PD perform worse cognitively than HCs, particularly in processing speed and verbal memory. Robust norming increases effect sizes and decreases PD scores to expected levels. The CSS outperformed individual tests and may detect cognitive changes in early PD, making it a useful outcome measure in clinical research. © 2025 The Author(s). *Movement Disorders* published by Wiley Periodicals LLC on behalf of International Parkinson and Movement Disorder Society.

Key Words: cognitive impairment; cognitive testing; neuropsychology; Parkinson's disease

[1]Department of Psychiatry and Neurology, Perelman School of Medicine at the University of Pennsylvania, Philadelphia, Pennsylvania, USA; [2]Parkinson's Research, Education and Clinical Center, Philadelphia Veterans Affairs Medical Center, Philadelphia, Pennsylvania, USA; [3]Department of Biostatistics, College of Public Health, University of Iowa, Iowa City, Iowa, USA; [4]Department of Neurology and Psychiatry and Behavioral Sciences, Baylor College of Medicine, Houston, Texas, USA

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

***Correspondence to:** Dr. Daniel Weintraub, Departments of Psychiatry and Neurology, Perelman School of Medicine at the University of Pennsylvania, 3615 Chestnut Street, Philadelphia, PA 19104, USA; E-mail: daniel.weintraub@pennmedicine.upenn.edu

Members of the Parkinson's Progression Markers Initiative are listed in the Supporting Information.

Relevant conflicts of interest/financial disclosures: D.W. has received research funding or support from The Michael J. Fox Foundation for Parkinson's Research, International Parkinson and Movement Disorder Society, National Institutes of Health, The Parkinson's Foundation, and the US Department of Veterans Affairs; honoraria for consultancy from Boehringer Ingelheim, Cerevel Therapeutics, CHDI Foundation, Citrus Health Group, Medscape, Modality.AI, Roche, Sage Therapeutics, Scion NeuroStim, and Signant Health; and license fee payments from the University of Pennsylvania for the Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease (QUIP) and Questionnaire for Impulsive Compulsive Disorders in Parkinson's Disease-Rating Scale (QUIPRS). M.K.Y. has received research funding or support from the Alzheimer's Association, the Michael J. Fox Foundation for Parkinson's Research, the Parkinson Foundation, and BlueRock Therapeutics.

Funding agency: This work was supported by The Michael J. Fox Foundation for Parkinson's Research (Research Grant "Creation and Performance of a Composite Cognitive Test Score in Parkinson Disease," MJFF-022598).

Received: 13 September 2024; **Revised:** 17 December 2024; **Accepted:** 23 December 2024

Published online 6 January 2025 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/mds.30111

Long-term significant cognitive impairment is a common and dreaded outcome in Parkinson's disease (PD), with dementia affecting up to 80% of patients long term.[1,2] Mild cognitive impairment (MCI) is also relatively common,[3] even in de novo or early disease.[4-6] In addition, subtle cognitive changes can be detected in the prodromal state, including in patients with isolated REM sleep behavior disorder (iRBD)[7] or persons with hyposmia.[8]

However, the ability to track cognitive changes over time, including in the context of short-term randomized controlled trials (RCTs), is hampered by consensus over which neuropsychological tests to use and how to use the results.[9,10] This may have contributed to having only one cognitive-enhancing medication US Food and Drug Administration approved for the treatment of PD dementia (rivastigmine)[11] with no other positive RCTs in the 20 years since then and no positive RCTs for PD-MCI.[12]

Increasingly, computerized cognitive batteries and smartphone-based cognitive apps are being used in PD, but most are not yet fully validated and ready for routine use in clinical research. Global screening instruments commonly used in PD (eg, Mini-Mental State Examination [MMSE][13] and Montreal Cognitive Assessment [MoCA])[14] are brief (ie, 5–10 minutes), but may not be sensitive to mild cognitive changes in the short term or may have significant variability.

Many individual paper-and-pencil cognitive tests are used in PD, both in clinical care and clinical research, and they are commonly assembled into a cognitive battery. Given the heterogeneity of cognitive deficits in PD, even at the stage of MCI,[3] batteries often assess multiple cognitive domains. Many versions of detailed cognitive batteries (eg, recommended Level I or Level II PD-MCI batteries[9]) are used, with some evidence that there is differential sensitivity among commonly used neuropsychological tests to mild cognitive deficits in PD.[15] Recent research has suggested that, in general, Level I and Level II cognitive batteries are better able to predict conversion from PD-MCI to PD dementia than are global screening instruments.[16] However, it remains unclear how to best use and interpret results containing multiple individual (sub)scores from differently scaled and normed tests.

An option that allows one to combine the richness and details of a neuropsychological battery with the simplicity of a single test score is to create a cognitive summary score (CSS) after all detailed test results are normed on the same scale (eg, *z* score). Examples from other neurodegenerative diseases include: (1) for Alzheimer's disease (AD), the Preclinical Alzheimer Cognitive Composite (PACC; consisting of total recall score from the Free and Cued Selective Reminding Test, delayed recall score on the Logical Memory IIa subtest from the Wechsler Memory Scale, Digit Symbol Substitution Test, and MMSE total score)[17] and the AD Composite Score (ADCOMS; four Alzheimer's Disease Assessment Scale–cognitive subscale items, two MMSE items, and all six Clinical Dementia Rating—Sum of Boxes items)[18]; and (2) for Huntington's disease (HD), the Huntington's Disease Cognitive Assessment Battery (HD-CAB; consisting of Symbol Digit Modalities Test, Paced Tapping, One Touch Stockings of Cambridge [abbreviated], Emotion Recognition, Trail Making B, and Hopkins Verbal Learning Test [HVLT]).[19]

In the Parkinson Associated Risk Syndrome (PARS) study, a CSS was generated from a very detailed cognitive battery and used to document subtle cognitive changes in prodromal PD (ie, persons with hyposmia + dopamine transporter [DAT] single-photon emission computed tomography [SPECT] scan deficit).[8,20] To our knowledge, there has not been other published literature documenting attempts to create a cognitive composite from detailed individual tests.

If determined to be more sensitive to early cognitive decline in PD, it is possible that a CSS would allow more neurobiological predictors of cognitive decline to be identified and even serve as an outcome measure for RCTs, including in persons with prodromal or at-risk PD. In addition, such a CSS could be valuable in the routine clinical evaluation of patients with PD, because clinical neuropsychological evaluations often administer the battery of standard neuropsychological tests used in PPMI.

The Parkinson's Progression Markers Initiative (PPMI) study originally had a PD-MCI Level I battery (five detailed, well-established cognitive tests) with published norms.[21] Given that PPMI PD participants are highly educated and motivated, we implemented robust norming (ie, removal of healthy control participants [HCs] who decline cognitively over time) to increase sensitivity to detect cognitive differences in PD compared with HCs, in a fashion similar to that employed to update the norms for the Dementia Rating Scale (DRS).[22] We hypothesized that using robust norming would increase sensitivity and adjust PD participant scores to the expected slight impairment reported in the literature,[6] and that a CSS would be more sensitive than individual test results in detecting cognitive differences in de novo, untreated PD patients compared with HCs.

Subjects and Methods

Standard Protocol Approvals, Registrations, and Patient Consents

An ethical standards committee on human experimentation reviewed and approved the study at each

site. Written informed consent was obtained from all participants in the study.

Cohort

The PPMI study and cohort have been extensively described.[23,24] All participants signed an approved informed consent form. Inclusion criteria for PD participants included (1) an asymmetric resting tremor or asymmetric bradykinesia, or at least two symptoms out of resting tremor, bradykinesia, and rigidity; (2) a recent clinical diagnosis of PD (mean [SD] duration from diagnosis = 8.5 [7.3] months); (3) being untreated with PD medications; (4) evidence of dopaminergic deficit based on DAT SPECT imaging; and (5) not having dementia.

HCs were required to (1) not have clinically significant neurological dysfunction and (2) not have a first-degree relative with PD. In addition, participants enrolled as HCs who had evidence of cerebrospinal fluid α-synuclein seed amplification assay positivity (n = 20) were not considered to be HCs for the purposes of this analysis.

Creating the Database

Baseline data were used, at which point PD participants were recently diagnosed and had not yet begun treatment with PD medication. The original cognitive battery was used to create the CSS to optimize the amount of data available for analyses. This original battery was composed of the HVLT-Revised (HVLT-R immediate and delayed free recall scores),[25] the Benton Judgment of Line Orientation 15-item version (JLO),[26] Symbol-Digit Modalities Test (SDMT),[27] Letter-Number Sequencing (LNS),[28] and semantic (animal) fluency.[29] Together these tests assess memory, visuospatial function, information processing speed, executive function, working memory, and language.

Published Norms

All cognitive tests used in the PPMI study are commonly used in clinical research (including in PD) and in clinical care, and have been extensively published using PPMI data.[4] Using previously published normative scoring methods for each test derived from the general population, and which correct variably for age and education, we derived external standard scores. This included T-scores for the HVLT-R (immediate and delayed free recall scores), SDMT, and semantic fluency, and scaled scores for the JLO and LNS.

Robust Norming

The criteria implemented to create the robust HCs were decided on by authorship group consensus, after reviewing similar efforts by other research groups[22] and reviewing changes in MoCA scores in PPMI HCs over the first few years of study participation. The application of the criteria is depicted in Figure 1. In brief, participants were required to have a baseline MoCA ≥ 27, have a year 1 MoCA score ≥ 26, and not have had more than a two-point drop in their MoCA score between baseline and year 1. This subset of participants was used to create internal norms for each test, which were then used to calculate internally derived *z* scores for each participant (see Statistical Analysis for details).

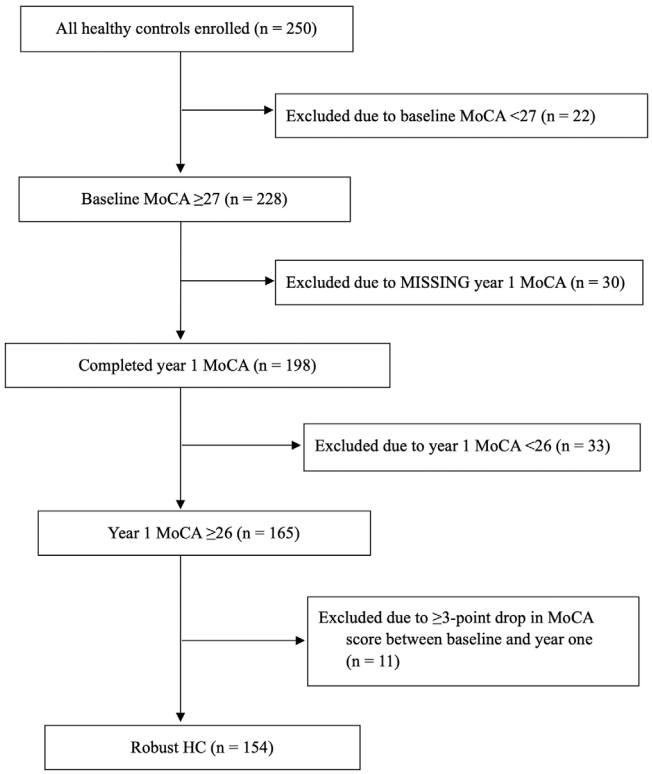


FIG. 1. CONSORT diagram for creation of robust healthy control (HC) group based on Montreal Cognitive Assessment (MoCA) performance.

Statistical Analysis

Statistical analyses were performed using SAS v9.4 (SAS Institute Inc., Cary, NC, USA; sas.com; RRID: SCR_008567). Statistical analysis codes used to perform the analyses in this article are shared on Zenodo (10.5281/zenodo.12636045).

All external standard scores derived from published norms (T-scores or scaled scores) were converted to *z* scores and then capped at −3 to 3. To derive internal norms, we used the robust HC population to build a linear regression model for each test. Age, sex, and continuous education were considered as possible predictors of raw test scores. The second-order polynomials of centered age and education were also considered. Sex was coded as a binomial variable with male = 1 and female = 2. Backward selection with a cutoff of

alpha = 0.10 was used to determine which of the demographic variables would appear in each final model. The Kolmogorov–Smirnov test was used to assess the normality of the residuals of each model. These derived linear models were used to calculate an expected raw score for each participant and each test, and then a z score was calculated by taking the participant's raw score minus their expected score and dividing by the root mean square error. These internal norms were also capped at −3 to 3.

Two CSSs were calculated for each participant, one being an average of their six external z scores (derived from published norms), and the other an average of their six internal z scores (derived from the PPMI robust HC subgroup), with the six z scores weighted equally in both cases.

For each variable in Table 1, two-sample t tests and χ^2 tests were performed to compare the PD group with each of the HC groups. Cohen's d effect sizes were used to compare these composite scores across groups.

Results

Cohort Characteristics

There were 923 participants with de novo, untreated PD; they had mean disease duration <1 year, were a mean age of 63 years, and were two-thirds male, highly educated, and overwhelmingly white and non-Hispanic (Table 1). There were 250 total HCs and 154 participants in the robust HC subgroup. PD participants were older than the total HCs (mean age, 61 years) and robust HCs (mean age, 59 years), but otherwise had similar demographic characteristics.

Raw Cognitive Test Scores

PD participants scored worse on global cognition (MoCA score) by approximately one point compared with both HC groups, as well as worse on all detailed cognitive test raw scores, except LNS for the entire HC group (Table 1). The entire HC group scored worse numerically on all cognitive tests compared with the robust HC subgroup (Table 1).

Using Robust HC Subgroup Instead of Entire HC Group

PD participants scored worse than the entire HC group and the robust HC subgroup on all detailed cognitive tests using both published norms (Table 2) and internally derived norms (Table 3). The effect size (Cohen's d) was greater (sometimes twice as great) when comparing the PD group with the robust HC subgroup rather than the entire HC group for all cognitive tests, whether using published norms (Table 2) or internally derived norms (Table 3). The tests with the biggest effect sizes, which were medium sized, were for information processing speed (SDMT) and verbal episodic learning and memory (HVLT-R). Smaller effect sizes were seen for JLO, LNS, and semantic fluency, but the effect size was increased for these tests by using the robust HC subgroup as the comparator group.

Regression-Based Norms Derived from Robust Healthy Control Participants

The results of regression analyses of the effects of age, sex, and education on the raw test scores of robust SH participants are summarized in Supporting Information Table S1. In brief, significant predictors included age and sex for the immediate and delayed recall scores of the HVLT-R; age, sex, and education for the JLO; age and education for the LNS; only sex for semantic fluency; and only age for the SDMT. For all tests where age was a significant predictor, older participants scored lower on average. Females scored significantly higher on the HVLT-R and semantic fluency, whereas males scored higher on the JLO. Participants with higher education performed better on the JLO and LNS. Kolmogorov–Smirnov tests (results not shown) indicated normally distributed residuals for HVLT-R immediate recall, LNS, SDMT, and semantic fluency; however, residuals were left skewed for HVLT-R delayed recall (skewness = −0.80; D = 0.09; $P < 0.01$) and JLO (skewness = −1.36; D = 0.11; $P < 0.01$).

Using Internally Derived Norms Instead of Published Norms

The effect sizes for PD versus both all HCs and the robust HC subgroup were similar for internally derived (Table 3) versus published norms (Table 2), except for semantic fluency, for which the effect size was approximately twice as great when using internally generated norms. The main change occurring with internally derived norms was a shift for nearly all tests (except HVLT-R delayed recall, which was relatively unchanged) to a more negative (ie, more impaired) z score, for both PD participants and HCs. Specifically, z scores for PD participants ranged from −0.50 to +0.65 using published norms, but −0.56 to −0.22 using internally derived norms.

Cognitive Summary Score

Using published norms, the CSS showed a larger effect size than all individual tests for both HC groups, except for the SDMT (Table 4). Using internally derived norms, the CSS had a larger effect size than all individual tests when using the robust HC subgroup as the comparator group (Table 4). Once again, changing from published to internally derived norms decreased the z score for the CSS from average (−0.01) to closer to low average (−0.40).

TABLE 1 *Demographic characteristics of PD and HC cohorts*

Variable	PD group	HC group		*P* value	
	All de novo PD (n = 923)	All HCs (n = 250)	Robust HCs (n = 154)	PD vs. all HCs	PD vs. robust HCs
Age at baseline, y				0.0077	<0.0001
Mean (SD)	63.1 (9.4)	61.2 (11.6)	59.1 (11.6)		
Median (min, max)	63.6 (33.5, 84.9)	62.8 (30.4, 85.3)	60.5 (30.6, 85.3)		
Sex, n (%)				0.2212	0.2548
Male	607 (65.8%)	154 (61.6%)	94 (61.0%)		
Female	316 (34.2%)	96 (38.4%)	60 (39.0%)		
Education, y				0.7029	0.8769
Mean (SD)	16.0 (3.1)	16.1 (3.1)	16.1 (3.0)		
Median (min, max)	16 (5, 32)	16 (6, 24)	16 (8, 24)		
Race, n (%)				0.0155	0.4051
White	856 (93.3%)	221 (88.8%)	140 (91.5%)		
Black or African American	18 (2.0%)	16 (6.4%)	8 (5.2%)		
Asian	14 (1.5%)	5 (2.0%)	2 (1.3%)		
Other	29 (3.2%)	7 (2.8%)	3 (2.0%)		
Ethnicity (Hispanic or Latino), n (%)	32 (3.5%)	4 (1.6%)	2 (1.3%)	0.1288	0.1523
Months since PD diagnosis				N/A	N/A
Mean (SD)	8.5 (7.3)	N/A	N/A		
Median (min, max)	5.9 (0.4, 42.8)	N/A	N/A		
MDS-UPDRS Part III				<0.0001	<0.0001
Mean (SD)	22.4 (9.7)	1.3 (2.1)	1.1 (1.8)		
Median (min, max)	21 (3, 62)	0 (0, 13)	0 (0, 10)		
MoCA				<0.0001	<0.0001
Mean (SD)	26.9 (2.4)	28.0 (1.5)	28.3 (1.1)		
Median (min, max)	27 (15, 30)	28 (20, 30)	28 (27, 30)		
HVLT-R immediate recall				<0.0001	<0.0001
Mean (SD)	24.3 (5.1)	25.8 (4.7)	27.0 (4.4)		
Median (min, max)	24 (9, 36)	26 (13, 36)	27.5 (14, 36)		
HVLT-R delayed recall				<0.0001	<0.0001
Mean (SD)	8.3 (2.7)	9.2 (2.4)	9.7 (2.2)		
Median (min, max)	9 (0, 12)	10 (2, 12)	10 (3, 12)		
Judgment of Line Orientation				0.0243	0.0079
Mean (SD)	12.8 (2.2)	13.1 (2.1)	13.3 (1.9)		
Median (min, max)	13 (1, 15)	14 (2, 15)	14 (4, 15)		
Letter-Number Sequencing				0.0607	0.0009
Mean (SD)	10.5 (2.7)	10.9 (2.7)	11.3 (2.8)		
Median (min, max)	11 (2, 20)	11 (2, 20)	11 (2, 20)		

(Continues)

TABLE 1 *Continued*

Variable	PD group	HC group		*P* value	
	All de novo PD (n = 923)	All HCs (n = 250)	Robust HCs (n = 154)	PD vs. all HCs	PD vs. robust HCs
Semantic fluency (animals)				0.0470	0.0018
Mean (SD)	21.2 (5.3)	21.9 (5.4)	22.6 (5.5)		
Median (min, max)	21 (7, 41)	21.5 (10, 38)	22 (10, 38)		
Symbol Digit Modalities Test				<0.0001	<0.0001
Mean (SD)	41.9 (10.3)	47.3 (10.3)	48.8 (10.5)		
Median (min, max)	42 (7, 82)	47 (20, 83)	48 (20, 83)		

Note: Report generated on data submitted as of November 6, 2023. Among de novo PD cohort, ≤10 participants missed any one assessment; among HCs, ≤2 participants in the overall group and ≤1 participant in the robust subgroup missed any one assessment.
Abbreviations: HC, healthy control participant; HVLT-R, Hopkins Verbal Learning Test-Revised; max, maximum; MDS-UPDRS, Movement Disorder Society Unified Parkinson's Disease Rating Scale; min, minimum; MoCA, Montreal Cognitive Assessment; N/A, Not applicable; PD, Parkinson's disease; SD, standard deviation.

TABLE 2 *Published norms* z *scores at baseline among de novo PD and HC cohorts*

Variable	PD group	HC group		Cohen's *d*[a]	
	All de novo PD (n = 923)	All HCs (n = 250)	Robust HCs (n = 154)	PD vs. all HCs	PD vs. robust HCs
HVLT-R immediate recall				−0.28	−0.47
Mean (SD)	−0.42 (1.09)	−0.12 (1.04)	0.08 (1.02)		
Median (min, max)	−0.40 (−3.00, 2.40)	0.00 (−3.00, 2.40)	0.10 (−3.00, 2.40)		
HVLT-R delayed recall				−0.34	−0.49
Mean (SD)	−0.50 (1.18)	−0.10 (1.09)	0.06 (1.01)		
Median (min, max)	−0.30 (−3.00, 1.40)	0.20 (−3.00, 1.40)	0.35 (−3.00, 1.40)		
Judgment of Line Orientation				−0.18	−0.25
Mean (SD)	0.65 (0.99)	0.83 (0.94)	0.89 (0.90)		
Median (min, max)	0.80 (−2.81, 2.49)	0.95 (−2.21, 2.57)	1.03 (−2.21, 2.57)		
Letter-Number Sequencing				−0.10	−0.20
Mean (SD)	0.52 (0.94)	0.61 (0.93)	0.71 (0.96)		
Median (min, max)	0.67 (−2.67, 3.00)	0.67 (−2.67, 3.00)	0.67 (−2.67, 3.00)		
Semantic fluency (animals)				−0.07	−0.14
Mean (SD)	0.10 (1.00)	0.17 (1.10)	0.25 (1.11)		
Median (min, max)	0.10 (−3.00, 3.00)	0.10 (−2.80, 3.00)	0.20 (−2.60, 3.00)		
Symbol Digit Modalities Test				−0.49	−0.60
Mean (SD)	−0.40 (0.98)	0.08 (1.02)	0.19 (1.03)		
Median (min, max)	−0.38 (−3.00, 3.00)	0.00 (−2.75, 2.90)	0.11 (−2.38, 2.90)		

Note: Report generated on data submitted as of November 6, 2023. Among de novo PD cohort, ≤13 participants were missing a *z* score for any one assessment; among HCs, 1 participant (not in the robust HC subgroup) had a missing *z* score for Judgment of Line Orientation.
Abbreviations: PD, Parkinson's disease; HC, healthy control; HVLT-R, Hopkins Verbal Learning Test-Revised; SD, standard deviation; min, minimum; max, maximum.
[a]Cohen's *d* was computed by dividing the mean difference between groups by the pooled SD.

TABLE 3 *Internally derived, regression-based* z *scores at baseline among de novo PD and HC cohorts*

Variable	PD group	HC group		Cohen's *d*[a]	
	All de novo PD (n = 923)	All HCs (n = 250)	Robust HCs (n = 154)	PD vs. all HCs	PD vs. robust HCs
HVLT-R immediate recall				−0.29	−0.52
Mean (SD)	−0.56 (1.11)	−0.25 (1.04)	0.00 (0.99)		
Median (min, max)	−0.52 (−3.00, 2.21)	−0.18 (−3.00, 2.37)	0.06 (−2.88, 2.37)		
HVLT-R delayed recall				−0.32	−0.48
Mean (SD)	−0.54 (1.14)	−0.17 (1.07)	0.00 (0.99)		
Median (min, max)	−0.33 (−3.00, 1.56)	0.03 (−3.00, 1.61)	0.18 (−3.00, 1.61)		
Judgment of Line Orientation					
Mean (SD)	−0.24 (1.07)	−0.04 (1.01)	0.01 (0.95)	−0.18	−0.23
Median (min, max)	−0.03 (−3.00, 1.80)	0.18 (−3.00, 2.02)	0.19 (−3.00, 2.02)		
Letter-Number Sequencing				−0.10	−0.22
Mean (SD)	−0.22 (0.95)	−0.12 (0.94)	0.00 (0.99)		
Median (min, max)	−0.20 (−3.00, 3.00)	−0.15 (−2.85, 3.00)	−0.10 (−2.85, 3.00)		
Semantic fluency (animals)				−0.13	−0.26
Mean (SD)	−0.25 (0.96)	−0.12 (0.98)	0.00 (1.00)		
Median (min, max)	−0.36 (−3.00, 3.00)	−0.18 (−2.31, 2.73)	−0.12 (−2.31, 2.73)		
Symbol Digit Modalities Test				−0.49	−0.55
Mean (SD)	−0.56 (1.01)	−0.07 (0.98)	0.00 (0.98)		
Median (min, max)	−0.56 (−3.00, 3.00)	−0.05 (−2.46, 3.00)	−0.04 (−2.46, 3.00)		

Note: Report generated on data submitted as of November 6, 2023. Among de novo PD cohort, ≤13 participants were missing a *z* score for any one assessment; among HCs, 1 participant (not in the robust HC subgroup) had a missing *z* score for Judgment of Line Orientation.
Abbreviations: PD, Parkinson's disease; HC, healthy control; HVLT-R, Hopkins Verbal Learning Test-Revised; SD, standard deviation; min, minimum; max, maximum.
[a]Cohen's *d* was computed by dividing the mean difference between groups by the pooled SD.

TABLE 4 *CSS using published and internally derived norms*

Variable	PD group	HC group		Cohen's *d*[a]	
	All de novo PD (n = 923)	All HCs (n = 250)	Robust HCs (n = 154)	PD vs. all HCs	PD vs. robust HCs
Published norms					
CSS				−0.42	−0.59
Mean (SD)	−0.01 (0.64)	0.25 (0.61)	0.36 (0.61)		
Median (min, max)	0.02 (−2.34, 1.99)	0.26 (−1.73, 1.99)	0.32 (−1.73, 1.99)		
Internally derived norms					
CSS				−0.41	−0.60
Mean (SD)	−0.40 (0.68)	−0.12 (0.62)	0.00 (0.61)		
Median (min, max)	−0.34 (−2.75, 1.76)	−0.12 (−1.85, 1.83)	−0.03 (−1.85, 1.83)		

Note: Report generated on data submitted as of November 6, 2023. Among de novo PD cohort, 17 participants had a missing value for both CSSs; among HCs, 1 participant (not in the robust HC subgroup) had a missing value for both scores.
Abbreviations: CSS, cognitive summary score; PD, Parkinson's disease; HC, healthy control participant; SD, standard deviation; min, minimum; max, maximum.
[a]Cohen's *d* was computed by dividing the mean difference between groups by the pooled SD.

Discussion

Using PPMI baseline cognitive data from de novo, untreated PD patients and demographically similar HCs, we demonstrated, as an illustrative example, that it is possible to create a CSS using robust norming that is sensitive to subtle cognitive changes in early PD.

Robust norming has been used to update the norms of commonly used cognitive tests,[30] given that studies have indicated that cross-sectional normative samples of older adults include a mix of cognitively normal individuals but also those in the initial stages of cognitive decline. This contaminates cross-sectional normative samples with cases of preclinical dementia that leads to an underestimation of the test mean and overestimation of the variance, thus reducing their clinical utility. When the DRS was renormed, use of the robust norms resulted in an almost twofold increase in sensitivity with relatively minimal loss of specificity when compared with the original norms.[22]

CSSs, also called cognitive composites, have been developed and used in clinical research for both AD and HD, including RCTs. The ADCOMS, developed to be sensitive to the early stages of AD and to changes with symptomatic therapy,[18] showed statistically significant improvement with lecanemab treatment in a recent AD trial.[31] The PACC has been shown to be sensitive to β-amyloid positivity in cognitively unimpaired individuals,[32] although it was negative as the primary outcome in a trial of solanezumab for preclinical AD.[33] For HD, the HD-CAB, which yields a composite *z* score that is sensitive in pre-HD and early HD groups and shows high test-retest reliability,[19] is recommended for use in clinical care and research, but may need modifications; also, there are no published results reporting on its use in an RCT.[34]

We found that patients with PD performed worse than HCs, whether using raw scores, published norms, or internally derived norms. This is consistent with what has been reported in other studies of newly diagnosed PD cohorts,[5] and with research showing that 15% to 30% of de novo patients meet criteria for MCI.[6]

Patients with PD also performed worse cognitively than HCs on all cognitive domains assessed, which is also consistent with multiple domains being impaired at the stage of MCI in PD[3] and makes the case that any cognitive battery in PD should assess multiple cognitive domains. Regarding specific cognitive tests, the most notable differences when comparing PD patients with HCs were for SDMT (information processing speed) and HVLT-R (verbal learning and memory), suggesting that these tests are good candidates for inclusion in a PD cognitive battery.

As part of robust norming, we excluded HCs who experienced cognitive decline over the first year of the study. Using the robust HC subgroup instead of the entire HC group significantly increased the effect sizes of cognitive differences between patients with PD and HCs, confirming the value in excluding HCs with incipient cognitive decline from the norming process.

The second stage of the robust norming process was the creation of internally derived norms instead of relying on published norms obtained from different populations, which is important given that the PPMI cohort, with both patients with PD and HCs, is highly educated (82% of the patients with PD reported having formal education beyond high school, and only two cognitive tests in the battery adjust for education), motivated, and younger than some other newly diagnosed cohorts. Using the internally derived versus published norms decreased the standardized test scores of patients with PD to expected levels (between average and low average) and allowed direct comparison across the tests because of the same reference population being used, as opposed to different reference populations for most published norms.

The final step after robust norming was to create a CSS by averaging the individual test *z* scores generated by robust norming. When doing this, and using the robust HC subgroup as the comparator, the CSS had a greater effect size than any of the individual tests comprising the battery, showing enhanced sensitivity for a summary score when using a robust HC group. This process also allows for applying *z* score cutoffs to define cognitive impairment, which might be used as an inclusion criterion or as an endpoint in clinical trials.

When trying to detect early cognitive changes in the context of a neurodegenerative disease with a high long-term incidence of dementia, ideally the change measured would be compared with a patient's premorbid performance, but these data are typically not available. Thus, we rely on comparing performance between patients and a normative group. However, the cognitive tests commonly used in neurodegenerative disease research have been normed in different populations with varying demographic characteristics and likely include individuals in early cognitive decline. To detect the mild cognitive changes present in de novo PD patients and given that the PPMI HC cohort is highly educated and motivated, we created a robust HC group for our cohort to norm all the cognitive tests, creating uniformity across them. We chose to use the global cognitive screening instrument that was not an instrument of our CSS (ie, the MoCA) to remove from our normative sample cohort all HC participants with evidence of short-term cognitive decline that might suggest the presence of the earliest stage of a cognitive disorder. By doing this, we substantially increased the effect sizes for differences in performance between PD and HC participants, saw an adjustment in average PD performance toward the expected (ie, mildly impaired

range), and made the tests directly comparable given the common normative cohort, thus achieving our goals. However, we acknowledge that although robust norming may increase sensitivity, it can be argued that it creates an artificial control population that does not represent the general community at large. Once the robust internal norming process was complete, the final step was the creation of the CSS by averaging the individual test z scores. Given that a range of cognitive domain deficits have been reported in PD, both in prodromal disease[8] and at the stage of MCI,[3] a similarly normed, equally weighted CSS would provide a single, easily interpretable score that covers the major cognitive domains affected in PD. We also envision that a CSS could be used as a primary outcome in intervention studies focused on cognition. Possible limitations of the CSS are the assumption that all cognitive domains are of equal importance, and that impairment in one domain and superior performance in another domain for a given individual will read out as normal performance on the CSS because of averaging.

Future research for a CSS in PD might include using other cognitive tests to compare performance against this battery, testing it against a global screening instrument (eg, the MoCA), testing it in prodromal PD (eg, iRBD or hyposmia), determining how the CSS evolves and predicts outcomes over time, determining its association with biological variables, and using it as an outcome measure in clinical trials. Future studies may also consider using optimized (ie, unequal) weighting of individual component test scores, such as those derived using linear mixed effects modeling, which could potentially optimize signal to noise and perform better than a simple equally weighted average.[35] However, the incorrect choice of weighting schema could lead to down-weighting an important item, or vice versa; also, weights that are optimal early in the disease course may become suboptimal as the disease advances longitudinally.[17,36] Several similar studies have decided against the use of unequal weights,[17,36] and we determined that unequal weighting was beyond the scope of this illustrative example and cross-sectional study design.

Strengths of the study include a very large sample size, a multisite international cohort, and use of well-validated and commonly used neuropsychological tests. One limitation is that the PPMI study does not administer a complete recommended Level II PD-MCI cognitive battery.[9] In addition, the PPMI cohort is not generalizable to the general or PD population at large, because it is a highly motivated research cohort, highly educated, and of very limited racial and ethnic diversity.

In conclusion, robust norming enhanced sensitivity in detecting cognitive differences in de novo, untreated PD and generated scores consistent with the mild cognitive deficits known to commonly occur in prodromal and early disease. The CSS generated after the robust norming process combines the richness of a detailed, multidomain cognitive battery with the simplicity of a single, identically normed and equally weighted score with potential for domain subscores. ■

Author Roles: Daniel Weintraub: (1) research project (a) conceptualization, (b) organization, and (c) execution and statistical analysis review and critique; and (2) manuscript (a) writing (first draft) and (b) review and critique. Michael C. Brumm: (1) research project (a) conceptualization, (b) organization, and (c) execution; and (2) statistical analysis (a) design, (b) execution, and (c) review and critique, and manuscript writing (first draft). Ryan Kurth: research project executions, statistical analysis execution and review, and manuscript review and critique. Michele K. York: (1) research project (a) conceptualization, (b) organization, and (c) executions and statistical analysis review and critique; and (2) manuscript review and critique.

Acknowledgments: PPMI, a public-private partnership, was funded by The Michael J. Fox Foundation for Parkinson's Research and funding partners, including 4D Pharma, AbbVie, AcureX, Allergan, Amathus Therapeutics, Aligning Science Across Parkinson's, AskBio, Avid Radiopharmaceuticals, BIAL, BioArctic, Biogen, Biohaven, BioLegend, BlueRock Therapeutics, Bristol-Myers Squibb, Calico Labs, Capsida Biotherapeutics, Celgene, Cerevel Therapeutics, Coave Therapeutics, DaCapo Brainscience, Denali, Edmond J. Safra Foundation, Eli Lilly, Gain Therapeutics, GE HealthCare, Genentech, GSK, Golub Capital, Handl Therapeutics, Insitro, Jazz Pharmaceuticals, Johnson & Johnson Innovative Medicine, Lundbeck, Merck, Meso Scale Discovery, Mission Therapeutics, Neurocrine Biosciences, Neuron23, Neuropore, Pfizer, Piramal, Prevail Therapeutics, Roche, Sanofi, Servier, Sun Pharma Advanced Research Company, Takeda, Teva, UCB, Vanqua Bio, Verily, Voyager Therapeutics, the Weston Family Foundation, and Yumanity Therapeutics.

Full Financial Disclosure for the Previous 12 Months: D.W. has received research funding or support from The Michael J. Fox Foundation for Parkinson's Research, International Parkinson and Movement Disorder Society,National Institutes of Health, The Parkinson's Foundation, and the US Department of Veterans Affairs; honoraria for consultancy from Boehringer Ingelheim, Cerevel Therapeutics, CHDI Foundation, Citrus Health Group, Medscape, Modality. AI, Roche, Sage Therapeutics, Scion Neuro Stim, and Signant Health; and license fee payments from the University of Pennsylvania for the Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease (QUIP) and Questionnaire for Impulsive Compulsive Disordersin Parkinson's Disease-Rating Scale (QUIPRS). M.K.Y. has received research funding or support from the Alzheimer's Association, the Michael J. Fox Foundation for Parkinson's Research, the Parkinson Foundation, and BlueRock Therapeutics.

Data Availability Statement

The data that support the findings of this study are openly available in Parkinson's Progression Markers Initiative (PPMI) at https://www.ppmi-info.org/access-data-specimens/download-data, reference number SCR 006431.

References

1. Aarsland D, Andersen K, Larsen J, Lolk A, Kragh-Sørensen P. Prevalence and characteristics of dementia in Parkinson disease: an 8-year prospective study. Arch Neurol 2003;60:387–392.
2. Hely M, Reid W, Adena M, Halliday G, Morris J. The Sydney multicenter study of Parkinson's disease: the inevitability of dementia at 20 years. Mov Disord 2008;23:837–844.
3. Aarsland D, Bronnick K, Williams-Gray C, et al. Mild cognitive impairment in Parkinson's disease: a multicenter pooled analysis. Neurology 2010;75:1062–1069.
4. Weintraub D, Simuni T, Caspell-Garcia C, et al. Cognitive performance and neuropsychiatric symptoms in early, untreated Parkinson's disease. Mov Disord 2015;30(7):919–927.

5. Aarsland D, Bronnick K, Larsen J, Tysnes O, Alves G, Group FtPS. Cognitive impairment in incident, untreated Parkinson disease: the Norwegian ParkWest study. Neurology 2009;72:1121–1126.
6. Weintraub D, Aarsland D, Biundo R, Dobkin R, Goldman J, Lewis S. Management of psychiatric and cognitive complications in Parkinson's disease. BMJ 2022;379:e068718.
7. Fantini ML, Farini E, Ortelli P, et al. Longitudinal study of cognitive function in idiopathic REM sleep behavior disorder. Sleep 2011; 34(5):619–625.
8. Chahine LM, Weintraub D, Hawkins KA, et al. Cognition in individuals at risk for Parkinson's: Parkinson associated risk syndrome (PARS) study findings. Mov Disord 2016;31(1):86–94.
9. Litvan I, Goldman J, Troster A, et al. Diagnostic criteria for mild cognitive impairment in Parkinson's disease: Movement Disorder Society task force guidelines. Mov Disord 2012;27(3):349–356.
10. Litvan I, Kieburtz K, Troster AI, Aarsland D. Strengths and challenges in conducting clinical trials in Parkinson's disease mild cognitive impairment. Mov Disord 2018;33(4):520–527.
11. Emre M, Aarsland D, Albanese A, et al. Rivastigmine for dementia associated with Parkinson's disease. N Engl J Med 2004;351: 2509–2518.
12. Seppi K, Ray Chaudhuri K, Coelho M, et al. Update on treatments for nonmotor symptoms of Parkinson's disease-an evidence-based medicine review. Mov Disord 2019;34:180–198.
13. Folstein M, Folstein S, McHugh P. "mini-mental state": a practical method for grading the cognitive state of patients for the clinician. J Psychiatr Res 1975;12:189–198.
14. Nasreddine Z, Phillips N, Bédirian V, et al. The Montreal cognitive assessment, MoCA: a brief screening tool for mild cognitive impairment. J Am Geriatr Soc 2005;53:695–699.
15. Hoogland J, van Wanrooij LL, Boel JA, et al. Detecting mild cognitive deficits in Parkinson's disease: comparison of neuropsychological tests. Mov Disord 2018;33:1750–1759.
16. Boel JA, de Bie RMA, Schmand BA, et al. Level I PD-MCI using global cognitive tests and the risk for Parkinson's disease dementia. Mov Disord Clin Pract 2022;9(4):479–483.
17. Donohue MC, Sperling RA, Salmon DP, et al. The preclinical Alzheimer cognitive composite: measuring amyloid-related decline. JAMA Neurol 2014;71(8):961–970.
18. Wang J, Logovinsky V, Hendrix SB, et al. ADCOMS: a composite clinical outcome for prodromal Alzheimer's disease trials. J Neurol Neurosurg Psychiatry 2016;87(9):993–999.
19. Stout J, Queller S, Baker K, et al. HD-CAB: a cognitive assessment battery for clinical trials in Huntington's disease. Mov Disord 2014; 29:1281–1288.
20. Weintraub D, Chahine L, Hawkins K, et al. Cognition and the course of prodromal Parkinson's disease. Mov Disord 2017;32: 1640–1645.
21. Wyman-Chick KA, Martin PK, Weintraub D, et al. Selection of normative group affects rates of mild cognitive impairment in Parkinson's disease. Mov Disord 2018;33(5):839–843.
22. Pedraza O, Lucas J, Smith G, Petersen R, Graff-Radford N, Ivnik R. Robust and expanded norms for the dementia rating scale. Arch Clin Neuropsychol 2010;25:347–358.
23. Marek K, Jennings D, Lasch S, et al. The Parkinson progression marker initiative (PPMI). Prog Neurobiol 2011;95:629–635.
24. Marek K, Chowdhury S, Siderowf A, et al. The Parkinson's progression markers initiative (PPMI) - establishing a PD biomarker cohort. Ann Clin Transl Neurol 2018;5(12):1460–1477.
25. Brandt J, Benedict R. The Hopkins Verbal Learning Test-Revised. Odessa, FL: Psychological Assessment Resources; 2001.
26. Benton A, Varney N, Hamsher K. Visuospatial judgment: a clinical test. Arch Neurol 1978;35:364–367.
27. Smith A. Symbol Digit Modalities Test: Manual. Los Angeles: Western Psychological Services; 1982.
28. Wechsler D. Wechsler Adult Intelligence Scale. 4th ed. San Antonio: Psychological Corporation; 2008.
29. Gladsjo J, Shuman C, Evans J, Peavy G, Miller S, Heaton R. Norms for letter and category fluency: demographic corrections for age, education, and ethnicity. Assessment 1999;6:147–178.
30. Holtzer R, Goldin Y, Zimmerman M, Katz M, Buschke H, Lipton RB. Robust norms for selected neuropsychological tests in older adults. Arch Clin Neuropsychol 2008;23(5):531–541.
31. van Dyck CH, Swanson CJ, Aisen P, et al. Lecanemab in early Alzheimer's disease. N Engl J Med 2023;388(1):9–21.
32. Papp KV, Rofael H, Veroff AE, et al. Sensitivity of the preclinical Alzheimer's cognitive composite (PACC), PACC5, and repeatable battery for neuropsychological status (RBANS) to amyloid status in preclinical Alzheimer's disease -Atabecestat phase 2b/3 EARLY clinical trial. J Prev Alzheimers Dis 2022;9(2):255–261.
33. Sperling RA, Donohue MC, Raman R, et al. Trial of solanezumab in preclinical Alzheimer's disease. N Engl J Med 2023;389(12): 1096–1107.
34. Considine CM, Rossetti MA, Anderson K, et al. Huntington study group's neuropsychology working group position on best practice recommendations for the clinical neuropsychological evaluation of patients with Huntington disease. Clin Neuropsychol 2024;38(4): 984–1006.
35. Ard MC, Raghavan N, Edland SD. Optimal composite scores for longitudinal clinical trials under the linear mixed effects model. Pharm Stat 2015;14(5):418–426.
36. Schneider LS, Goldberg TE. Composite cognitive and functional measures for early stage Alzheimer's disease trials. Alzheimers Dement (Amst) 2020;12(1):e12017.

Supporting Data

Additional Supporting Information may be found in the online version of this article at the publisher's web-site.